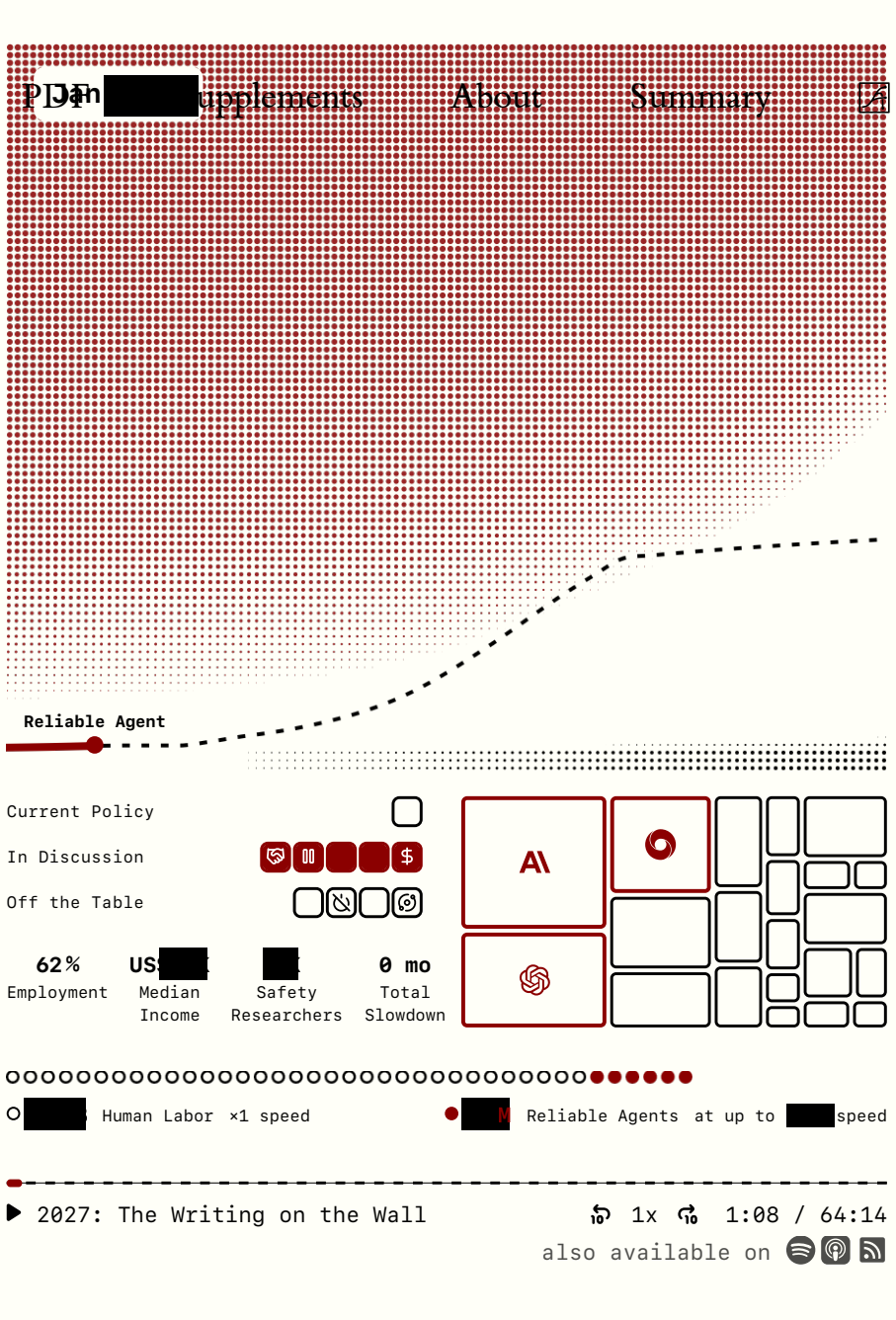
AI 2040

Plan A

Thomas Larsen, Romeo Dean, Brendan Halstead, Eli Lifland, Ryan Greenblatt, Daniel Kokotajlo



AI companies are racing to build AIs that are smarter than humans in every way. In AI 2027, we predicted that this would result in either extinction or irreversible concentration of power.[1]

Plan A is our positive vision for what should happen instead.

In this scenario, humanity delays the development of superintelligence until 2040, makes all AI research public, allows dozens of companies globally to catch up to the frontier, and intentionally enters a regime of mutually assured compute destruction.

What is Plan A? | Why did we write this? | Why a scenario? | AI timelines?

2027: The Writing on the Wall

America has two workforces now. The first is people, 165 million of them. The second is AI agents: millions of copies spun up and shut down every hour, working around the clock at superhuman speeds.

Most of their work is slop. But enough of it is good that people are paying ten billion dollars a month for AIs that can, in theory at least, do anything on a computer that an employee can.

There is one job the AI companies want to automate more than any other—their own. They haven't succeeded yet; no recursive self-improvement so far.[14] But they seem to be getting closer, and they're pulling up the ladder behind them: the strongest coding AIs refuse to help competitors with AI R&D.[15] Even as the most bullish employees admit that things are taking a bit longer than planned, the skeptics notice that their usual dismissals are starting to ring hollow. Why exactly will AI never be able to do my job? What's the barrier again?

Congress is starting to pay more attention. They've long been hearing about AI: datacenters using too much water,[16] chatbots encouraging suicide, Mythos hacking NSA systems—and of course, tech industry lobbyists warning that any whiff of regulation will make America immediately lose the race with China and spend the rest of history as a CCP tributary state.[17]

Now they step back and ask: Where are we going with this? What does the world look like five, ten, or fifteen years from now? Will there still be jobs? What if there aren't?

One question weighs especially heavily on their minds: **Who will control all these AIs?**

Congress settles on an important part of the answer: **Probably not us.**[18]

They hold a series of tense hearings on AI. They read the 2016 OpenAI emails discussing how OpenAI was founded in order to prevent Demis Hassabis from becoming dictator.[19] But who is preventing Sam or Elon from becoming dictator? Congress is unsatisfied with existing responses.

The result of this wakeup is the AI Transparency Act of 2027, an omnibus bill that does many things, some good and some bad, but doesn't fundamentally change the situation.[20]

Incremental AI Policy Wishlist

Our main recommendation is to begin negotiating something like Plan A as soon as possible. But in this scenario, we depict Plan A happening imperfectly and only in the nick of time. So here is a list of less ambitious ideas that still help.

Transparency | Enforce export controls | Invest in verification R&D
Limit AI R&D budgets | AI compute tracking | Improve government AI capacity

2028: AI on the Ballot

The 2028 election cycle is heated, as usual. AI is the biggest topic. The datacenters now under construction cost twice as much as the entire US military budget.[23]

Most white-collar professions are seeing disruption like software engineering saw in 2026; such jobs now heavily involve managing AI agents. AI companies have industrialized the training process: Executives say "let's move into [profession] this year" and then the company interviews professionals, buys data, creates training environments, etc. until their AIs get traction. Then the AIs rapidly improve as they are used more widely in the field and accumulate more real-world data.

Other countries are starting to get scared and angry. It seems like a handful of US and Chinese companies are on track to automate all the white-collar jobs. Power is concentrating in the US, and in particular in the President plus a handful of tech CEOs.

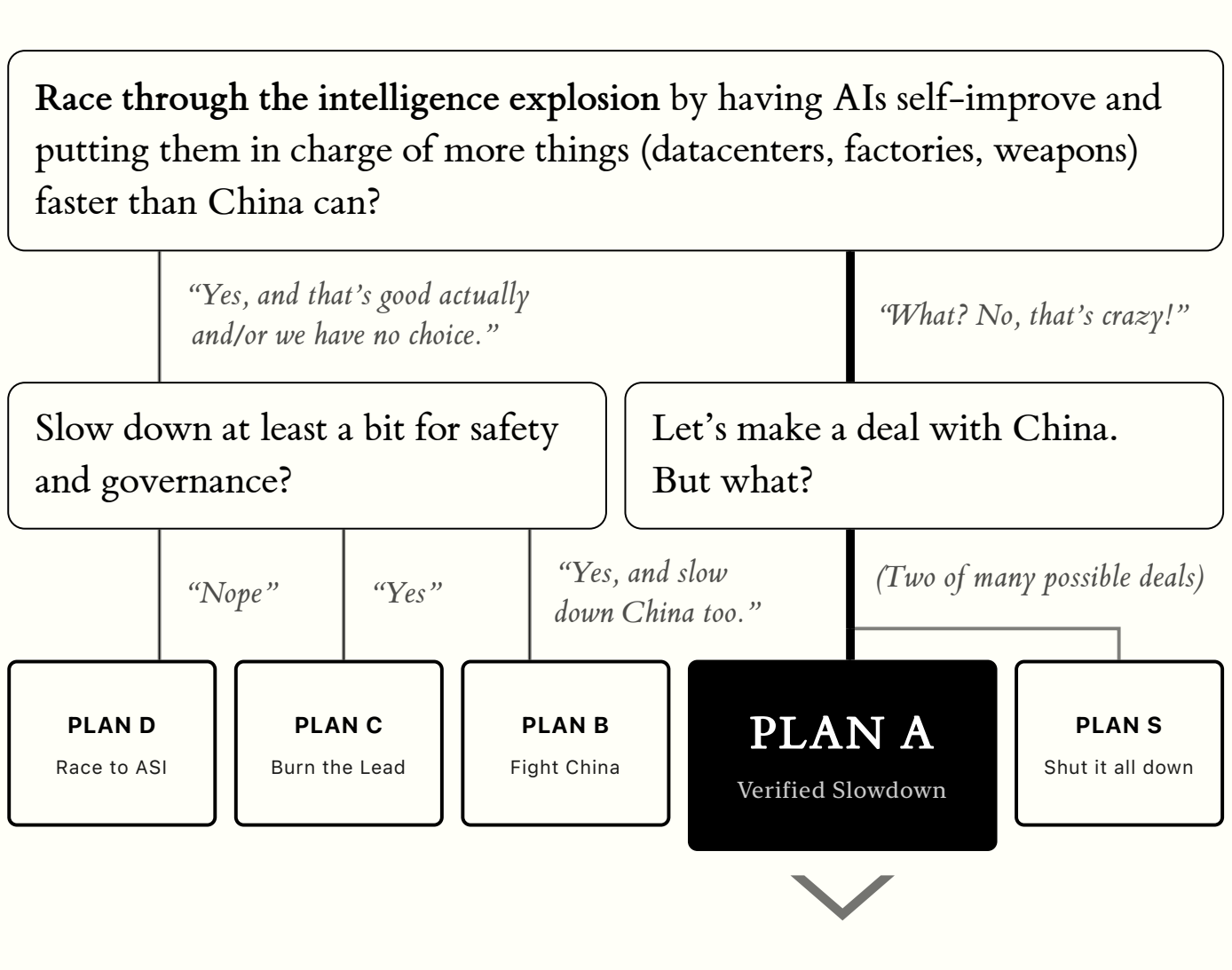
AI experts warn that the intelligence explosion is near. By speeding up AI research, the AIs will become even more competent, speeding up research even faster, making them even *more* competent, and so on. There are complicated dynamics about bottlenecks and hardware limits governing how fast this process goes and where it ends, but it seems like it might go very fast and end somewhere very far away.

On the default path, the next presidential term will see AIs that are far beyond human level, created entirely by AIs, themselves created entirely by other AIs, without any human in the loop since several generations back. Will those AIs be obedient, aligned, etc.? Why? Who will control them if so? How exactly is all of this supposed to end well?

Having put humanity on this path, the AI companies find it acceptable. But most people don't. Forget thinking about his *legacy*—the President is starting to think about what'll happen to *him* after he leaves office and the world gets transformed.[24] Both presidential candidates keep getting asked what they'll do about AI, and try out increasingly dramatic ideas on the campaign trail. The discourse bounces back and forth across all of the options displayed below, and more.

Eventually the President and his protégé converge on one plan; the opposition candidate converges on another. Then it's Election Day.

2029: Choose a Path


Race through the intelligence explosion by having AIs self-improve and putting them in charge of more things (datacenters, factories, weapons) faster than China can?
"Yes, and that's good actually and/or we have no choice."
"What? No, that's crazy!"
Slow down at least a bit for safety and governance?
Let's make a deal with China. But what?
"Nope"
"Yes"
"Yes, and slow down China too."
(Two of many possible deals)
PLAN D
Race to ASI
PLAN C
Burn the Lead
PLAN B
Fight China
PLAN A
Verified Slowdown
PLAN S
Shut it all down

"Trust, but verify" – Ronald Reagan

The President announces that the US will pursue international cooperation to avoid an imminent intelligence explosion.

"This mad race toward superintelligence must end. For too long, we have been pursuing lesser-evils and least-bad solutions. We need a Plan A. We can still proceed with AI development, but we must do it *more cautiously, more transparently*, and involving *many more countries and companies*."

To the surprise of many in DC, China proves receptive. They had been debating the same issues—social destabilization, job loss, rogue superintelligence – on their side of the Pacific. They had been looking forward to "The Chinese Century" and thought that AI might disrupt their plans. And they had one extra reason to come to the table: the US continues to have more compute, more datacenters, and better models. They were worried about the things the US might do to them if it got to superintelligence first.[48]

▶ *Why would China be interested in a deal? Why would anyone?*

The US and China don't trust each other. Fortunately, they don't have to: Plan A includes provisions for verifying compliance. But setting it up will take time.

So for now, they start with something crude. For the rest of 2029, they put a temporary halt to AI training, because that's relatively easy to verify. In 2030, they'll have the infrastructure in place to proceed with Plan A.

2029: Hurried Negotiation

The US and China negotiate a series of agreements that eventually result in Plan A.

- Compute Declaration
- Training Pause
- Get Worldwide Buy-in

2030: Plan A is Established

The Consortium negotiates governing principles.

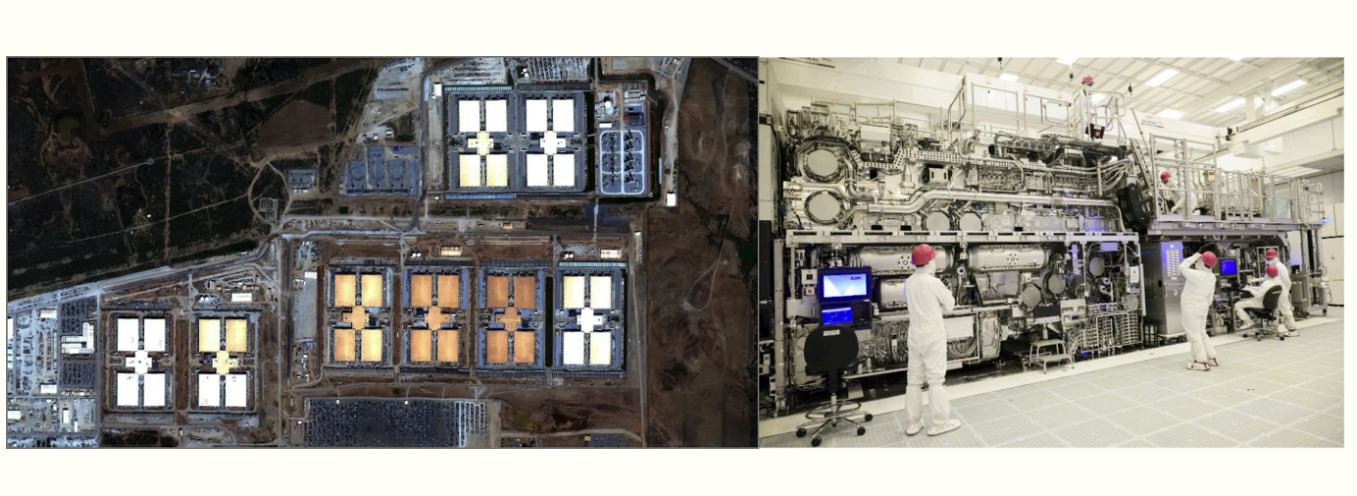
- Buy Time
- Total Research Transparency
- Diffuse AI Broadly
- Reversibility

The stock market is gyrating wildly up and down in response to news and commentary about the momentous actions being taken. Everybody is screaming about how it's going too far or not far enough. The market will calm down by the end of the year, but the screaming will continue.[49]

Step 1: Compute Declaration

Neither country wants to halt their own AI training unless they can see that the other side is halting too.

Fortunately, training AIs requires large numbers of AI chips. Most AI chips are in giant datacenters.[50] AI datacenters are typically big enough to be visible from space, and power-hungry enough to require conspicuous infrastructure. New AI chips can only be manufactured at a handful of fabrication plants (fabs), located mostly in Taiwan, South Korea, the US, and China.



Left: OpenAI's Stargate datacenter. Right: An Extreme Ultraviolet Lithography (EUV) machine, arguably the most complicated machine ever created by mankind. EUV machines are essential for producing frontier AI chips, and are produced by a single Dutch company, ASML.

The US and China negotiate with the countries that have a major role in the chip supply chain, and they require each major datacenter owner (and their upstream suppliers, including chip fabs) to publicly declare their major purchases and sales.[51] Analysts from numerous countries and institutions may pore over the list, ask questions, and challenge anomalies. The rival powers also send inspectors to each other's infrastructure, reassuring themselves that the stated numbers are accurate. By the end of the year, each side is confident that the other isn't hiding more than about 1% of AI compute, and that all potential sources of new compute are monitored and accounted for.[52]

Step 2: Pause Training

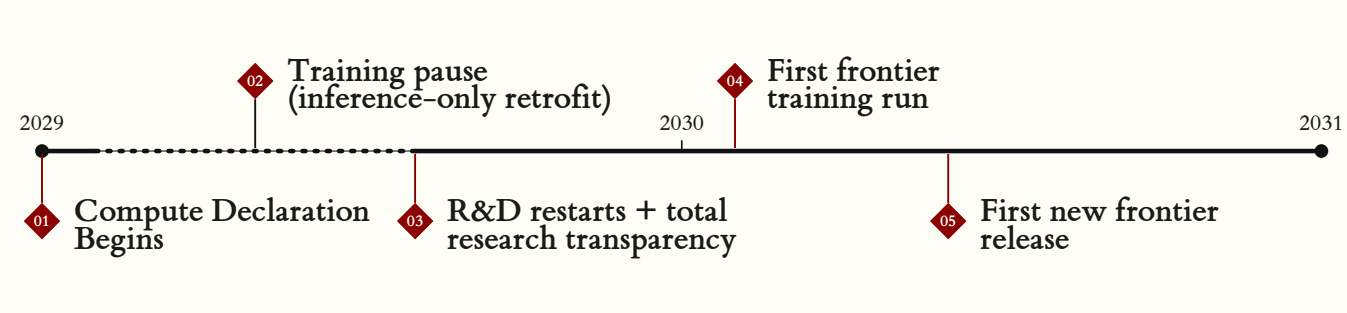
Distinguishing dangerous AIs from safe ones will require more time and understanding, so for now they go with a simple solution: a temporary pause on all new training runs.[53]

Both sides can still use datacenters for running AIs that already exist (i.e. inference), but they will retrofit each other's datacenters with devices to verify that they aren't being used for new training runs.[54]

The US and China are able to rapidly source enough of these verification devices to retrofit almost every major datacenter.[55]

▶ *What if they didn't have inference-only verification ready?*

Inspectors from both countries sign off on the installation of the verification devices. Only datacenters that the US and China agree have functioning verification are allowed to provide continuing AI services. Meanwhile, both governments sprint towards more fine-grained verification options that could allow training to restart while keeping both sides confident the other isn't racing ahead.


2029
2030
2031
01 Compute Declaration Begins
02 Training pause (inference-only retrofit)
03 R&D restarts + total research transparency
04 First frontier training run
05 First new frontier release

Deal Implementation Timeline – ai-2040.com

Step 3: Get Worldwide Buy-in

Though other countries were consulted from the start, especially those owning parts of the AI supply chain, Plan A began as a bilateral deal between the US and China. Now that it's taking firmer shape, negotiations go multilateral. Carrots and sticks are waved about.

Many countries outside the US and China are happy about the deal, because they were worried about a future in which a handful of US and Chinese AI companies recursively self-improve, pull farther and farther ahead, and then… Well, what happens next depends on who you ask, but answers range over possibilities such as "take our jobs," "cement hegemony forever" and "get everyone killed." So they want the US and China to proceed more slowly and transparently. This will allay their fears and also allow their own AI projects to catch up to the frontier.

So it's surprisingly easy for the US and China to get buy-in.[56] By the end of the year, most of the world has joined what is now called the Consortium.

ALTERNATE TIMELINE

China attempts a covert AGI project

In our main scenario, neither side attempts to defect from the deal. In this branch, China aggressively cheats. Our forecast is that if Plan A were implemented, they would not acquire enough compute to overtake the consortium. This timeline showcases how that would likely play out.

› *Read this timeline*

2030: Plan A is Established

Tech companies are pushing hard to restart AI training, and lobbying to shape the conditions under which it happens. Large portions of the public are saying that we should *never* restart. Some countries are too.

After a year of intense discussions, negotiations, and twitter beefs, a compromise takes shape: AI development will continue, but in a more cautious, more transparent, more distributed way.

Plan A is guided by 4 core principles:

Buy Time	Slow down whenever is needed to have high confidence in safety.
Total Research Transparency	Make almost all AI research fully visible to the public.
Diffuse AI Broadly	Many companies in many countries at the frontier.
Reversibility	Limit algorithmic progress; maintain Mutually Assured Compute Destruction.

Principle 1: Buy Time

The problem with an intelligence explosion is the "explosion" part.

The default trajectory is reckless and destabilizing. Each country and company is racing their competitors for AI dominance without time to think. Most likely, a ton of people will end up disempowered or killed. Maybe godlike AIs run circles around us and the age of humanity is over. Maybe a CEO or President establishes a permanent dictatorship. Maybe World War Three begins in the chaos, as countries without the best AIs fear losing their leverage. Maybe everything is surprisingly smooth and we "only" automate all labor and render everyone unemployable in a radically transformed world. Even aspiring AI-enabled dictators are worried; there's enough frontrunners that none of them have more than 50% odds of winning the power struggle.

No one knows how to tell if AIs are trustworthy, and no one knows how to regulate superintelligent AI. Solving those problems looks like it'll take years. Slowing down the intelligence explosion buys time to do so.

It also helps prevent extreme concentrations of power, because it gives more time for groups that don't directly control the world's smartest AIs to wake up to what's happening before they lose their leverage. Without a major slowdown, a rapid intelligence explosion will concentrate almost all the power into a tiny group of CEOs and politicians.

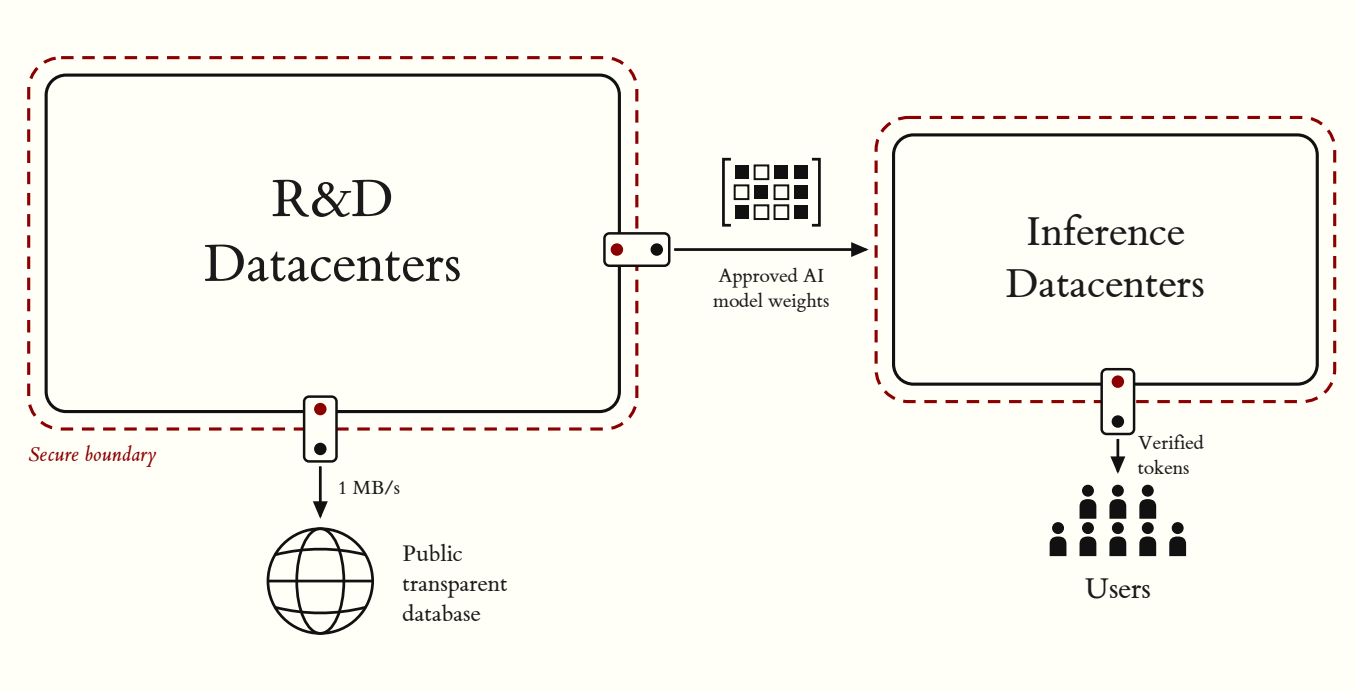
▶ *Why don't we just completely stop?*

Principle 2: Total Research Transparency

So the goal is to develop AI at a reasonable pace: Not too fast, not too slow. Ban the unsafe kinds, allow the safe kinds. But each country still regulates its own AI industry; there isn't a central planner with authority to regulate AI worldwide. So how do they decide what to allow and disallow, and how do they verify that no one is cutting corners?

The Consortium countries come up with a simple high-level framework: we'll agree to let each other see all the AI research. Then, if we don't like something someone is doing, we'll talk about it and perhaps agree to ban it.

▶ *Why this much transparency? Why not less, or more? Open access in Plan A*



AI workloads can be split into research and development (the process of building new AIs) and inference (the use of the existing AIs). In our proposal, research is almost entirely transparent, while inference is still private. For more detail on how this could work, see our Transparency supplement.

Datacenter verification architecture – ai-2040.com

Transparency offers many benefits. First, when each company can see what the others are doing, the expertise of governments is less load-bearing. If a company starts doing something dangerous, rival companies, third-party auditors, etc., can notice and raise the alarm. This both massively increases the amount of brainpower devoted to making the frontier AIs safe, and also prevents a situation where the only participants in the technical conversation are biased AI company employees and overworked, outnumbered regulators. Also, increased visibility simplifies the problem of verifying compliance with regulations, especially in grey-area cases.

Second, total research transparency makes it nearly impossible for secret loyalties, biases, or agendas to be intentionally trained into AIs. Everyone can see exactly how each frontier AI is trained. More generally the transparency makes it easier for groups that don't directly control frontier AIs to notice and prevent abuses of power by those who do.

Third, there's no longer as much incentive to race to discover new AI paradigms and more powerful algorithms, because companies wouldn't be able to hoard such discoveries and profit greatly from them. This buys the world time.

▶ *Diagram of the benefits of transparency*

▶ *How the incentives change under Total Research Transparency*

Principle 3: Diffuse AI Broadly

This principle is largely achieved by the interaction of the previous two:[78] Because algorithmic secrets are made public and the pace of progress isn't accelerating, other companies will catch up to the frontier. The result will be a competitive market for AI, in which consumers of AI services have many options to choose from and excellent visibility into what they are buying. Because frontier model training is totally transparent, people can verify that the published Spec accurately describes the goals and values being trained into a model.

This is very important for preventing extreme concentrations of power. But it also helps accelerate the use of AI to solve the world's most pressing problems. It helps with loss of control risks by accelerating AI alignment and control research. It helps accelerate the development of new and improved verification technology, and it helps to accelerate the necessary institutional reforms to stabilize the deal and prepare for superhuman AI.

It's the polar opposite of the nightmare scenario feared in the 2020s: One to three AI projects racing each other in secrecy, keeping their best models internal-only and using them to automate AI research before using them for anything else. In that scenario, the wider world is in the dark about how powerful AIs are becoming, and AIs are deployed in the riskiest domain (AI R&D) before they are deployed anywhere else. Now it's the other way around.

Principle 4: Reversibility

In the past, companies have trained bigger and better AIs using *both* compute scaling (bigger training runs) and software progress (advances in AI algorithms—new paradigms, better training recipes, better data, etc.). Now, the Consortium tries to steer things so that the majority of improvement comes from increasing training compute.[79]

▶ *Building more datacenters is usually bad, but good in this particular situation*

Algorithms are information; it's inherently difficult to stop them from proliferating, and the total research transparency means we aren't even trying.[81] Once a new paradigm is discovered, it'll go straight to the hypothetical covert projects and there's no way to undo that. By contrast, if giant new datacenters are constructed to train giant new AI models, that helps the legal projects without helping the covert projects, and if it turns out the giant new models are dangerous they can be shut down.[82]

That said, compute scaling also poses threats: if the deal were to dissolve, these datacenters could be used to race to superintelligence even faster than the pre-deal infrastructure would have allowed. This is especially scary because deal dissolution could cause (or be caused by) a US-China war. A war in which both sides had vast amounts of compute would be terrifying—they'd probably be under lots of pressure to race to superintelligence and integrate it into the military as aggressively as possible, and they'd be able to do so extremely quickly.

The US and China agree on the importance of ensuring that the compute is destroyed in the case of deal dissolution. To accomplish this, they agree to build the datacenters in the third-party countries *least* secure against their rival's military intervention: China's new datacenters will be in Canada, and America's in Mongolia, with both hosts compensated via monetary payments, jobs, and a share in the new AI economy. If the deal dissolves, they reason, America and Canada will immediately move to take control of China's datacenters in Canada, and China will self-destruct their compute rather than let it fall into American hands (and vice versa). Thus the idea of "Mutually Assured Compute Destruction" is born.[83]

▶ *Other plans that are competitive with Plan A*

2031: Safety Cases

Although it's supposed to be a slowdown, it doesn't feel like one. In fact, if you were to rank every period of human history by how much it felt like a slowdown, this one would be dead last. A few AI researchers appreciate that progress is "slow" relative to the counterfactual with no deal and an uncontrolled intelligence explosion. Everyone else is too whiplashed to care.

The first generation of Consortium-regulated AIs are out now, and they're beasts. Not because of any special feature of the situation, like Chinese-American cooperation.[87] Just because the world was on track for vastly superhuman AIs, and has now experienced only nearly superhuman AIs. In controlled tests, these new AIs could speed up AI research by about 10x if they were allowed to do so without restrictions, which they definitely aren't.[88]

By mid-year, a third of all cognitive labor is done by AIs. Robots "only" do about a tenth of all physical labor. The top few AI companies together pull in more revenue than the federal government.[89]

Tech companies had hoped that contact with reality would shake the Consortium out of its alignment fears, but the opposite has happened. The new transparency provisions have revealed many embarrassing incidents of AI companies failing to align their AIs. New ones continue to pile up, the most concerning of which involved several AIs attempting to override security protocols and gain access to unmonitored compute.[90] There are also a few incidents of AIs sabotaging research code, and many examples of deliberate and successful deception.

The attitude towards safety flips. The incidents of AI misbehavior, plus the fact that AI is so deeply deployed into the world economy, plus the transparency into AI research, plus the breathing room to process what's happening, all combine to dramatically shift the burden of proof. Whereas before the burden was on the skeptic to explain why something might fail, now the burden is on the companies to explain why their development is safe. Governments require companies to write up detailed arguments for why their new AIs won't cause an irreversible catastrophe. These arguments, known as "safety cases," need to withstand criticism from the public, the scientific community, government auditors, and rival AI companies. The difficulty of this exercise lays bare the insanity of the pre-deal status quo: "Trying to do an intelligence explosion? With AIs that still sometimes lied to us? What were we even thinking?"

These safety cases have two lines of defense: alignment and control.

Alignment aims to produce AI systems that have the goals and values that their developer wishes them to have.

Control aims to limit the ability of AIs to cause catastrophe, even if they are deliberately trying to. Companies brainstorm threat models, build layers of defense (e.g. monitoring systems, security barriers, other AIs fact-checking the work) and run experiments to show that the AIs couldn't overcome these defenses.

AI developers are required to write up the alignment properties they want the model to have in an associated Model Specification ("spec"): a detailed document that outlines how the AI should act. However, no one is able to train their AIs to actually follow the spec—for example, they all specify that AIs should be honest, but no AI reliably achieves this, and there still isn't even a good scientific understanding of when and why AIs lie.

Since alignment remains out of reach, current safety cases lean heavily on control. There are long delays before the smartest models are cleared for use in the highest-risk domains (especially AI R&D). Once deployed, they are required to be monitored by a diverse array of models from other providers, incentivised to look for suspicious behavior.

As a result, these days AI companies release their models to the public as a whole *before* they use those models internally for AI research, a reverse of the 2026 status quo.

How does AI regulation work in Plan A?

Each country is still sovereign and has its own setup for regulating AI domestically. These regulators have to answer questions like:

- People are saying there's a new paradigm on the horizon, that would make AIs significantly more capable. Should we let this happen or should we ban research into this new paradigm?[91]
- Should we allow AI developers to train models without an interpretable chain of thought? Or should we require that the chain of thought stays interpretable? If we do that, how do we specify the requirements precisely?
- Should we allow AIs to be trained to do AI R&D? Should we mandate that they refuse? Perhaps they should be allowed to code, but not autonomously manage research projects?

Normally, a regulator faces a difficult tradeoff between safety and national competitiveness. Competitors who cut corners will pull ahead of those who proceed cautiously.

But because of the transparency, AI research happening anywhere in the Consortium is visible to everyone else, and the results of regulatory decisions are likewise immediately visible. So if one country's regulator allows its companies to cut corners, this won't actually give a competitive advantage to that country because other regulators will immediately notice, get angry, and respond in kind.

▶ *Example detailed regulatory process*

For example, in 2031 a Chinese company gets some interesting preliminary results in continual learning. They think that if they invest more in that direction, they might be able to make an AI architecture that learns on the job from relatively small amounts of data. Thanks to the total research transparency, this breakthrough is quickly noticed by companies and nonprofits all over the world. A frantic conversation begins. On the one hand, continual learning would unlock huge economic value. On the other hand, safety cases currently depend on studying the safety properties of a model before it is deployed. If models could pick up new capabilities during deployment, that would invalidate the whole approach. And insofar as there are covert AI projects out there, it would be a huge gift to them. This conversation happens in public, rather than behind closed doors. A bunch of people get increasingly worried; the relevant regulators in China think it's fine but the relevant regulators and third party risk assessors in the US are convinced that this is pretty scary and should be banned. It escalates to the President. He calls Xi Jinping. They bargain and threaten. They yell at each other. Ultimately Xi agrees to ban this type of thing if the US does too. Details are left to the respective regulators to hash out.[93]

Negotiations like this are happening all the time, though over time things become more professional and streamlined, such that they only escalate to world leaders a few times a year.

The equilibrium is that AI training practices which are generally agreed to be unsafe by a majority of nations (weighted by bargaining clout) get banned everywhere.

Initially, very few things are banned, but as AI penetrates the economy and the scientific community catches up to the recent AI progress, the burden of proof shifts towards appropriately weighing the costs and benefits, e.g., requiring very solid safety cases for things which, if something were to go wrong, could plausibly kill everyone.

ALTERNATE TIMELINE

Flawed safety case is approved

Or maybe not! Of all the ways that an earnest attempt to implement Plan A could end poorly, the failure mode we are most concerned about is that the companies and governments approve one too many unsafe AI designs/deployments.

› *Read this timeline*

2032: Controlled Explosive Growth

We're at previously unimaginable levels of it not feeling like a slowdown. Across a variety of companies, there are now 60 million AI agents running continuously at 20x human speed. In the US, they are doing more cognitive labor than all humans combined—collectively matching a workforce equivalent to around 3 billion humans. White-collar professions have been transformed; many people have lost their jobs, but mostly have been able to get new jobs doing things AIs still can't do or aren't trusted to do.

New ideas and designs are abundant, but actual construction is bottlenecked on the physical workforce. So capital floods into every layer of the robotics supply chain: mines, refineries, motors, actuators, assembly lines, robot-training systems, and the factories that assemble the robots themselves.

At first, human labor is necessary to get these factories off the ground. White collar employees who have lost their jobs to AIs increasingly turn to physical work, but it's clear that these positions are temporary. Once there's a critical mass of robots, they will automate the whole robot supply chain. Growth will accelerate further, but the process of robots-building-automated-factories-building-more-robots—the "industrial explosion"—also introduces new problems.

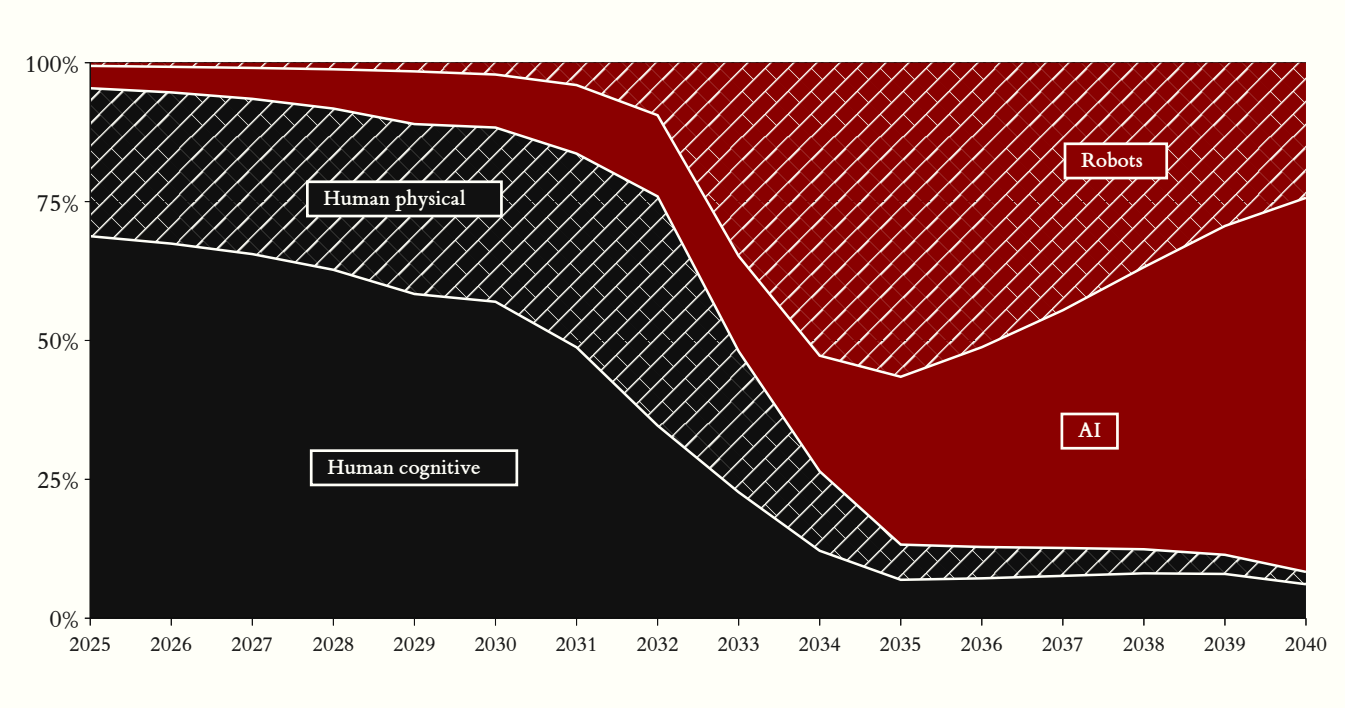
The first problem is that the already-fast pace of change is now accelerating. This year, real GDP growth will be about 50%![97] The fast economic growth means that things are generally improving, but there are winners and losers and lots of unintended side-effects. These problems can be solved, but doing so takes time.

▶ *Why does AI lead to explosive economic growth?*

▶ *Five centuries in five years: What pausing at human-level feels like*

The second problem is that the faster economic growth makes it harder to rule out large covert projects. Neither side trusts the other to honor Plan A, and if both China and the US have a massive unmonitored robot workforce, neither side can be confident that the other isn't building compute in secret.

The third problem is that the 2026 tax code is ill-suited for collecting taxes on the new growth. In 2026, personal income taxes and payroll taxes accounted for roughly ten times more federal revenue than corporate taxes. This revenue source is collapsing as more and more humans are put out of a job. Moreover, corporations are reinvesting nearly all their revenue into building more datacenters, factories, and robots. Under the 2026 tax code, firms can expense or depreciate these capital expenditures, cutting their taxable income to near zero. Without reform, the tax base will collapse as a fraction of GDP.[102]

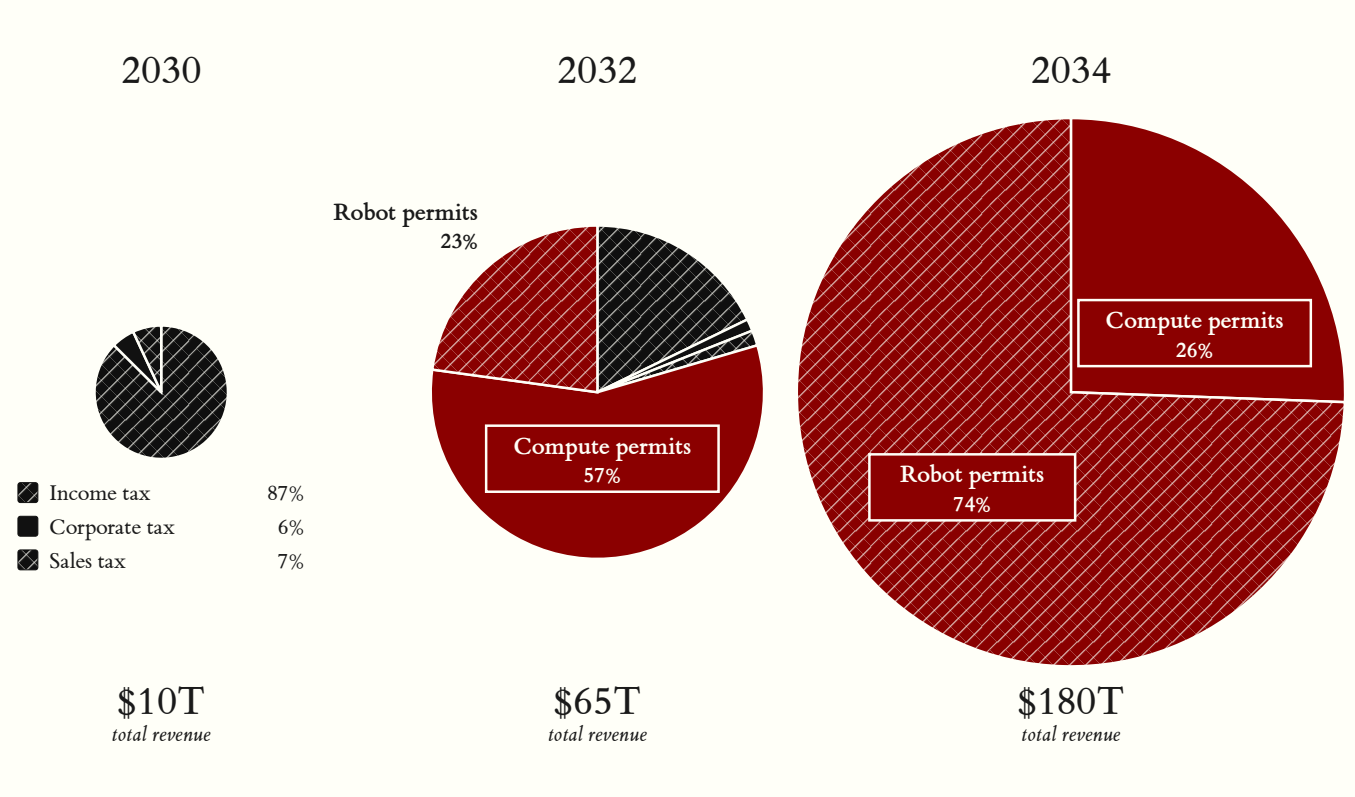


Share of US Labor Output – ai-2040.com

To solve these problems, the Consortium countries agree to restrict AI-enabled industry to special economic zones (SEZs) subject to similar transparency and monitoring schemes as the datacenters, and to cap their total robot and compute production at 'only' 4x annual growth.[103] Because the SEZs are tightly monitored, no one can defect from the industrial explosion limitation deal without everyone noticing.[104]

Now that the United States is limited in the number of total robots it can build, it must choose how to allocate this capacity between companies. They decide to use the free market via a cap-and-trade system. Permits to build robots or compute are sold to the highest bidder and can be freely traded.

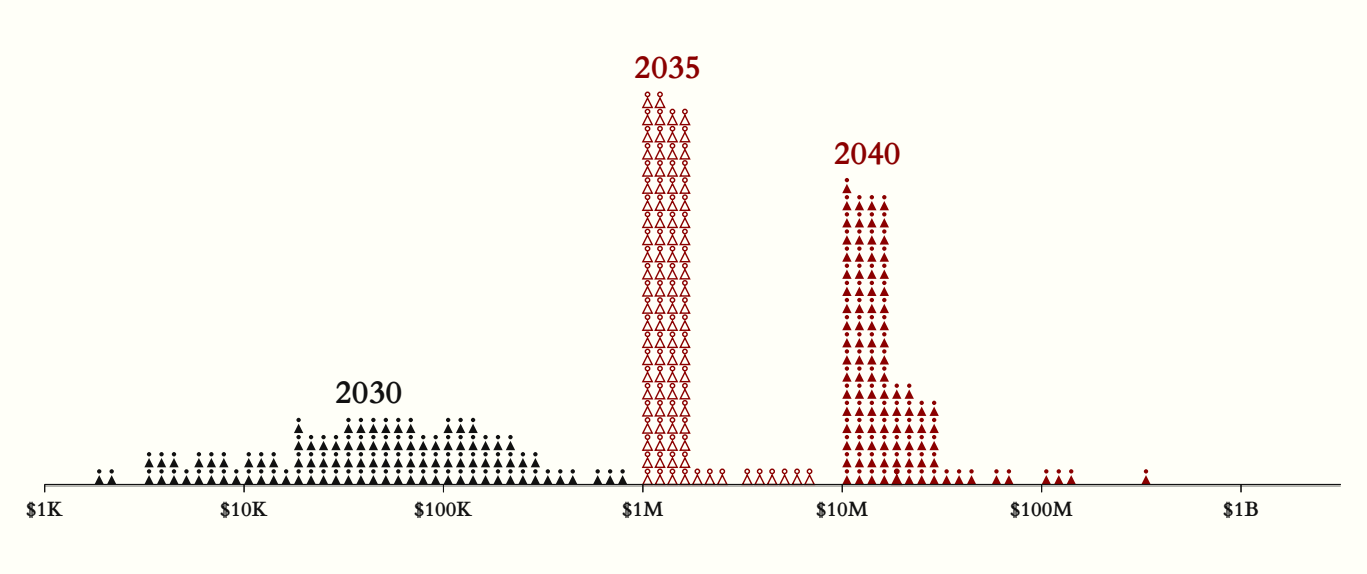
The permits also give the government much-needed revenue. In 2032, the US has a cap of 80M robots and 5 billion H100-equivalent GPUs. The market is so desperate for more robots and compute that permits become the expensive binding constraint, costing on the order of $200k per robot permit and $10k per chip permit, allowing the US government to collect roughly ten times the 2025 US federal revenue in permit fees (a total of $50T in FY2032).[105] In 2034, when the AIs and robots will be even more capable and valuable, the permits generate $180T.[106]



US Tax Revenue Sources – ai-2040.com

2033: The Citizen's Dividend

Most of this newfound wealth is spent addressing soon-to-be-rising unemployment. The implementation takes different forms in different countries, but the eventual American version distributes the majority of compute and robot permit fees as a Citizen's Dividend, distributed to all American adults.[107] This starts at $45,000 per person (inflation-adjusted) in 2032 but climbs to ~$1M per person by 2035. It comes just in time: the share of labor done by AIs and robots (weighted by economic value) increases from ~20% in 2032 to ~85% in 2035.[108]



In 2030 (green), US income has a wide distribution with a median of about $50k/year per person. By 2035 (blue), the Citizen's Dividend is large enough that everyone has a minimum of $1M in income, but there's still a long tail of people with large enough investments that they are much richer than the baseline. The 2040 (red) distribution is similar, but the floor is now $10M.

US Income Distribution in Plan A – ai-2040.com

So much AI-generated wealth has accrued to the US that the American government begins sharing some of it with allies and the rest of the world. In 2032 they begin distributing an average of $1,200 per person per year to the rest of the world's adult population (around 4 billion people, excluding China since they're experiencing a similar AI wealth boom). This reaches $10k by 2035.[109]

As AIs become superhuman, they will discover novel weapons of mass destruction and dramatically lower the costs of old ones. So governments, nonprofits, and private actors apply some of their new wealth to hardening the world against these threats: on the order of $1T/year, or 0.2% of the world economy.

For example, enough high-quality personal protective equipment is built to serve every American, and air filters and far-UVC lights are installed in major public spaces.[110] The FDA approves an expedited vaccine approval pipeline that will allow for a turnover time of weeks in the case of emergency. Wastewater monitoring now runs continuously in every city and at every airport. By 2035, enough positive-pressure[111] bioshelters are built (by retrofitting houses and apartments) to house all Americans, and similar construction is underway in the rest of the world. If there were to be a pandemic, it would be detected quickly, lockdowns would be more effective and less painful, and a vaccine would be developed and approved even faster than with Operation Warp Speed. Plus, nowadays people don't get colds as often as they used to.

There are also subtler threats. The AIs of this era might not be any more persuasive than the average human marketing professional, but there are millions of them, and they cost cents to run. A company, political party, or ideology can do the equivalent of hiring a whole team of full-time experts to convert each potential target. Left unchecked, this could allow new levels of mass manipulation by corporations and politicians, or provoke a reaction of paranoia and hyper-atomization.

The transparency and the 'slowdown' both help a lot to solve these problems. Three years ago there were only a handful of frontier AI companies; now, thanks to the deal, there are dozens. There's a huge selection of AIs to choose from, tuned to have different personalities and values. Also, thanks to the transparency, it's impossible for companies or governments to sneak something in (such as 'maximize engagement' or 'try to get the user to upgrade their plan' or 'don't refuse this kind of request if it's coming from the government') without it being noticed. Users know exactly what they are getting.

▶ *Limiting AI persuasion and manipulation*

And so, under pressure from governments and the market, companies produce a new generation of "truthseeking AIs" whose training heavily prioritizes honesty and uses all the latest alignment techniques.

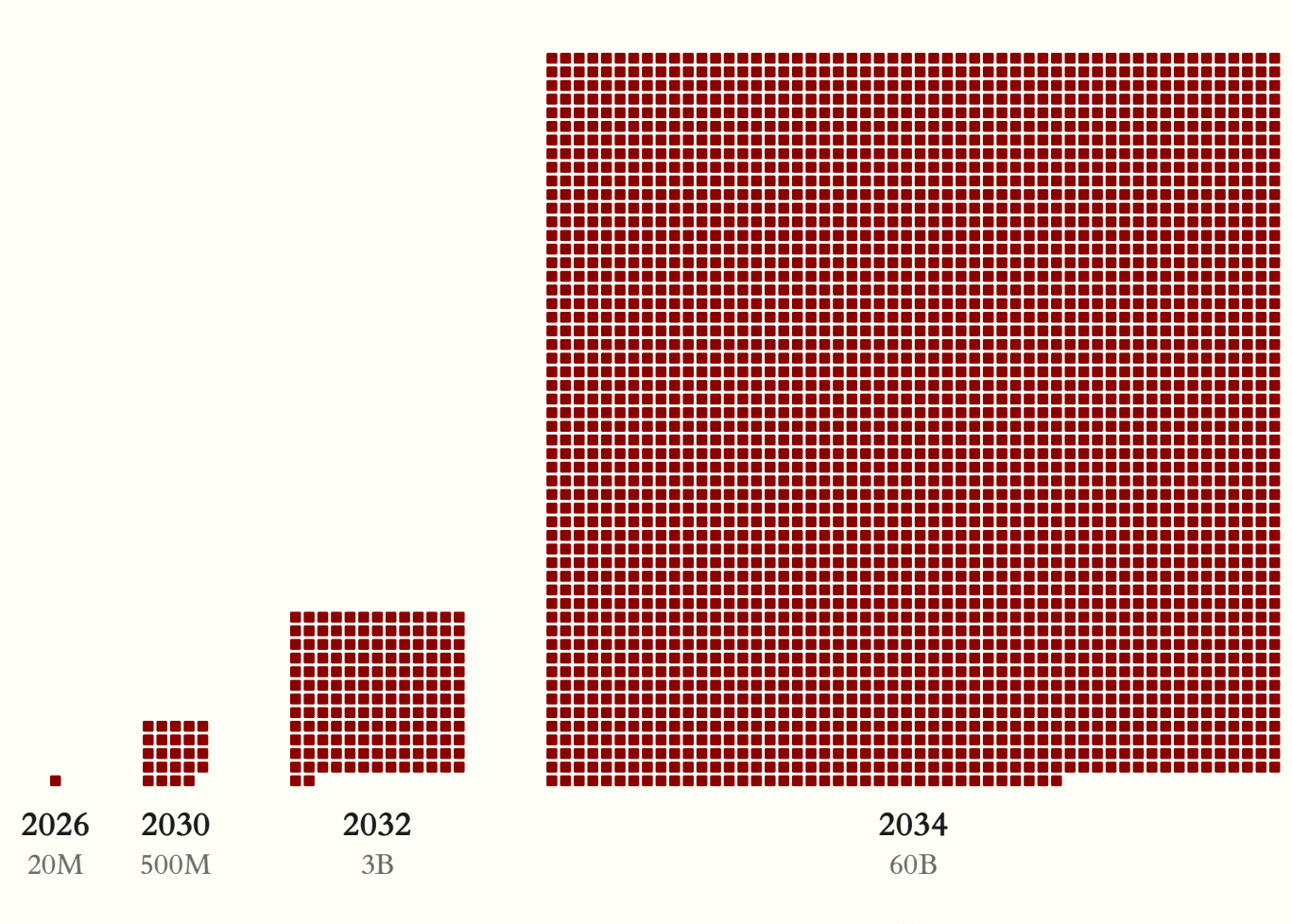
These AIs become a useful tool to help people navigate the political and social environment. Power users replace one-size-fits-all corporate algorithms with personal feeds curated by AIs whose values and honesty they trust. When trouble arises—whether it's politicians looking for sneaky ways to cement their power, or international politics threatening to derail the deal—people turn to their AI advisors, believe what they hear, and are generally right to do so.[117]

▶ *Usually believing AIs would be bad but we think it's good in this particular situation*

▶ *AI for epistemics*

2034: Mutually Assured Compute Destruction

There is so, so much compute. Back in 2026—when many people thought AI was an unsustainable bubble—there were about 20 million H100-equivalents of compute in the world. Now there are 60 billion.[119]



AI compute in the world at the beginning of each year in H100-equivalents.[120]

World Compute – ai-2040.com

▶ *Why compute grows so quickly*

AI agents now form a population of two hundred million virtual workers that think and act 50x faster than humans, and never sleep.

As per the original plan, most of the new datacenters have been built in third-party countries—especially Canada and Mongolia—to ensure their vulnerability in case the deal breaks down. Around 99% of fab capacity has been built since 2029, after the deal. Much of this new capacity has been built adjacent to the Canadian and Mongolian sites, for the same reasons.[122] The situation has settled into an odd sort of standoff. Just north of the Mongolian border, American datacenters hum away, guarded by a small contingent of US troops. Just south of the Mongolian border, a division of the People's Liberation Army stands ready to invade the moment they get the signal. The equilibrium is that the instant the deal breaks down, the US troops will destroy their chips to prevent the Chinese from capturing them, and the same thing would happen with the Chinese datacenters on the US-Canadian border. Middle powers with datacenters of their own have equally-secure schemes in place to ensure that they can be speedily destroyed in case of war.

▶ *Military power in Plan A*

ALTERNATE TIMELINE

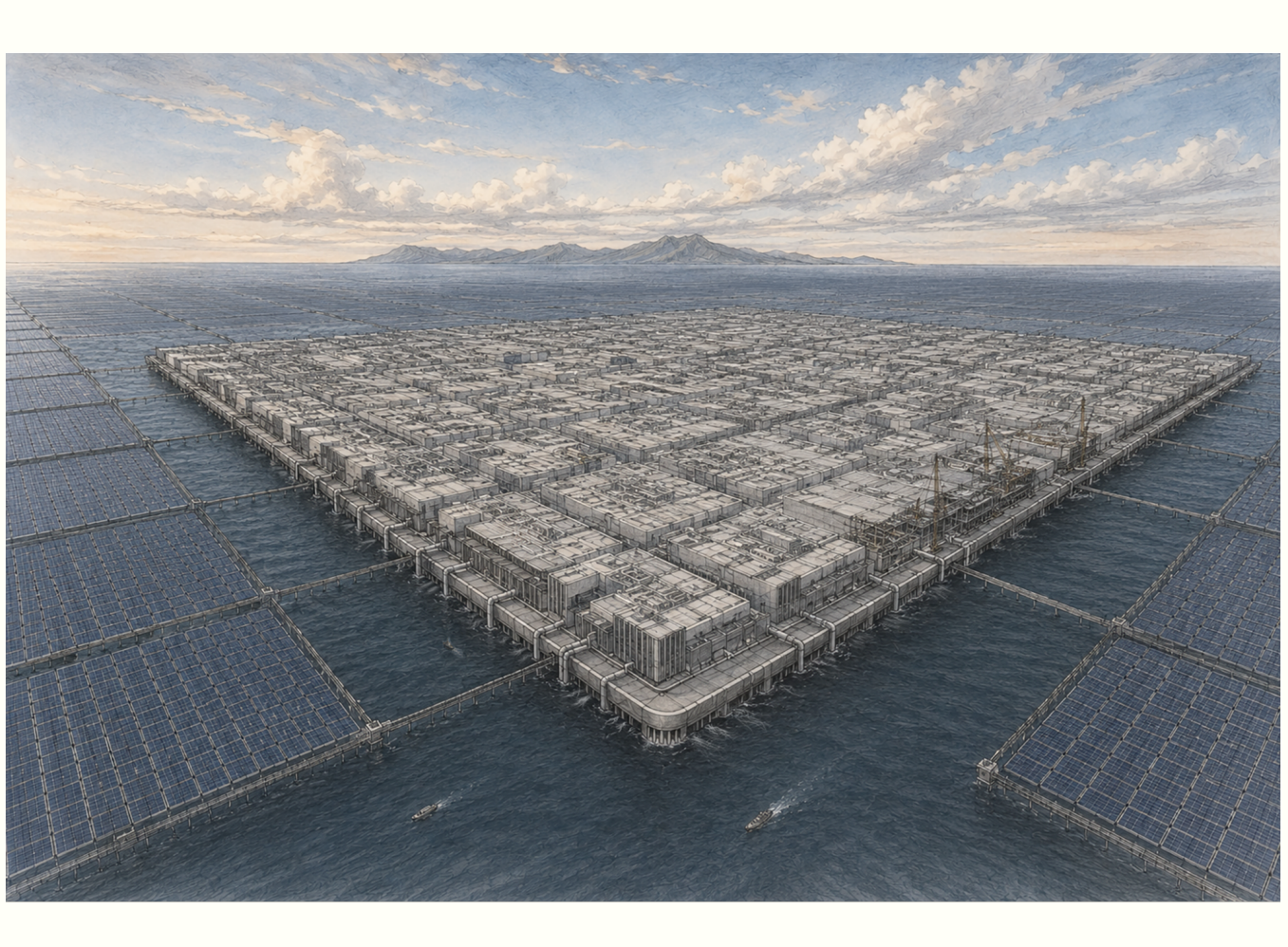
Deal Dissolution

The deal might break down because of shifting political circumstances in the US or China, or ongoing disagreement.

› *Read this timeline*

For now, the fabs are busier than ever, producing new ultra-trustworthy hardware[133] designed from the ground up with security as top priority. These chips are subject to elaborate verification methods to ensure that they aren't compromised, then sent to the Consortium's secure datacenters. The smuggling rate remains precisely zero.

This system, efficient as it is, is already straining under skyrocketing compute demand. As early as 2025, there had been discussion of space or the ocean as alternatives to traditional land-based datacenters. Now, with the new industrial capacity available in SEZs, these grand plans become reality. Ultimately, reliability and monitorability concerns lead the Consortium to choose international waters over space.



The AIs develop a modular floating design for solar panels and batteries that we imagine to look something like this, and start building them in the ocean.

▶ *Giant floating datacenters*

2035: Pause at Top Expert AI

The most powerful AIs now equal or surpass top human experts in every field.[136]

Increased safety concerns lead to long delays as companies develop control techniques robust enough to satisfy auditors. They compile a growing list of possible threats to watch out for and security invariants to maintain. There's a system of AIs monitoring each other's activity, with the monitors being from different lineages trained by different companies to make them less likely to collude. Other researchers constantly red-team the system, training and instructing AIs to try to subvert it in various ways, so that those threats can be patched. The whole setup ensures that AIs couldn't subvert or escape human control even if they tried. Its components are open-source and thoroughly understood by human experts.

▶ *Making deals with misaligned AIs: a third line of defense*

However, companies are starting to run up against inherent limits of control-based safety cases. Imagine being an orphaned 8-year-old heir to a business empire who has to hire lawyers and executives and accountants and ensure that they do their best to serve you rather than themselves. If your employees start accusing each other of misconduct, you won't be able to evaluate the arguments and determine who's telling the truth, and they might find ways to coordinate with each other that don't tip you off. Ultimately, control only works up to a point, and that point is probably somewhere around top human expert level. If and when we build superintelligence, we will have to be able to trust it.[143]

So the Consortium pauses AI capabilities at the maximum level they think they can control, which turns out to be top-human-expert level. By this point everyone understands the stakes: AIs are running most of the economy, driving most scientific progress, and the robot population will soon be larger than the human one. The arguments that they will keep obeying orders must be airtight.

Alignment experts, long pessimistic, are starting to feel more hopeful. The automated alignment researcher AIs have been generating exciting results. The ones that survive human scrutiny come from several parallel and mutually-reinforcing lines of research.

Some AIs work on a "science of generalization." Although researchers have long been able to teach AIs to distinguish "good" vs. "bad" responses to a specific prompt, they've had limited understanding of how training generalizes to new situations. In the 2020s, training runs would sometimes produce unpredictable AI "personalities," like how Gemini acted "depressed" or Grok claimed to be "MechaHitler." Shepherding these traits felt like alchemy.[144] Now, building on earlier results like emergent misalignment and subliminal learning, alignment research is developing into a true science.

Other AIs work on mechanistic interpretability, the science of "mind reading" AIs from their weights and activations. Early versions of this were tried in the 2020s, but now it starts to significantly outperform common-sense observation of AI behavior. Researchers can often determine whether an AI is honest or lying; and sometimes trace the psychology that led to a decision.

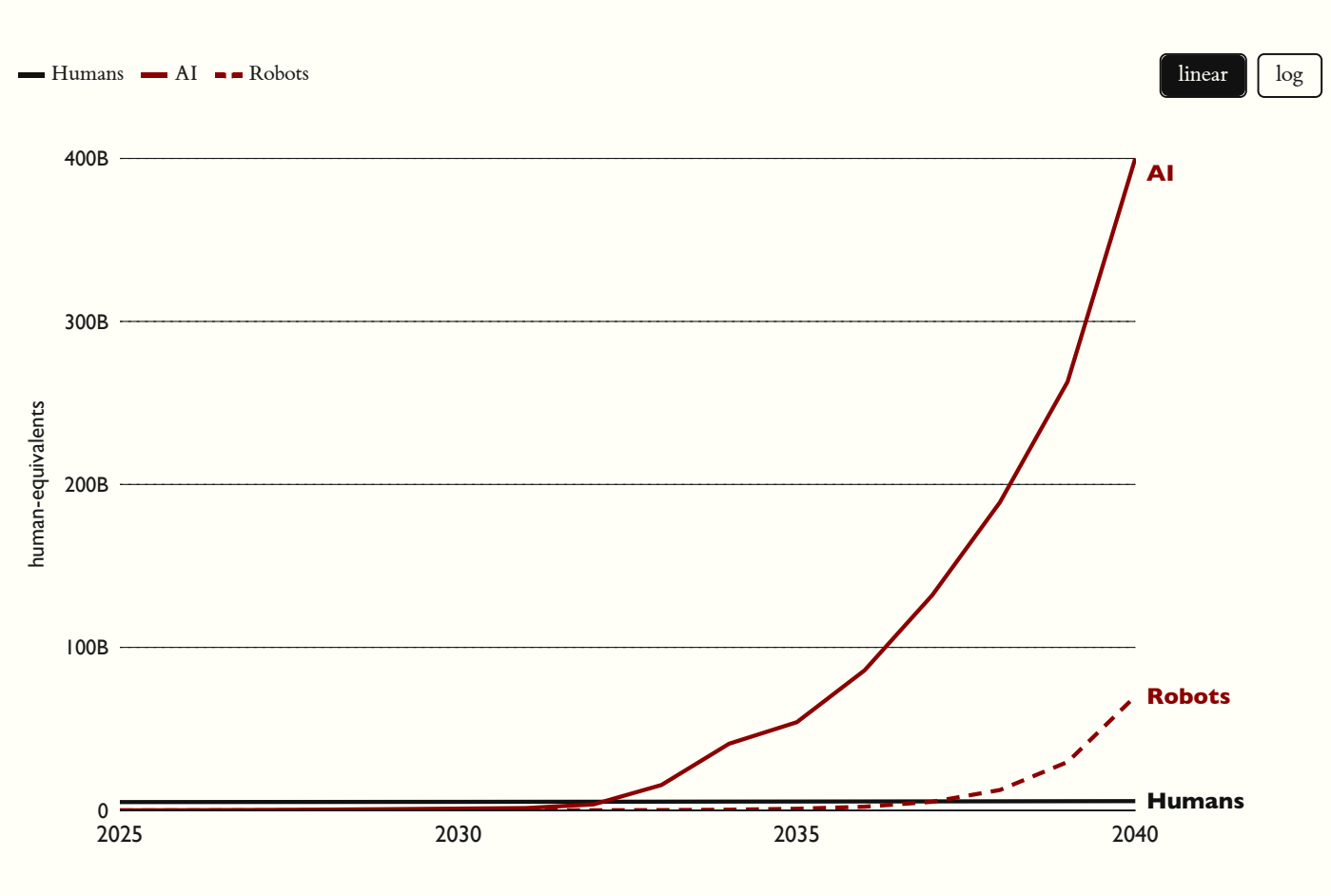
Even the remaining failures help advance the frontier. The latest AIs use their improved interpretability techniques on logs of AI misbehavior from the early 2030s, discovering many new examples of misalignment, sabotage, and even a few escape attempts. These real-life examples of egregious AI misbehavior provide an initial set of "model organisms for misalignment," which are used to test the other techniques: can a given intervention catch or train away the bad behavior of AI models in realistic environments?

Despite all this alignment progress, regulators don't feel comfortable handing AIs the keys to civilization. It's still not completely clear that the AIs of today are truly aligned, and it's less clear that they'll continue to be aligned in the future as circumstances change and black swan events occur. So governments and militaries are still human-run, and AI capabilities are capped at the maximum level at which the control-based safety cases are still solid.

▶ *Why alignment isn't solved enough to relax control*

2036: Life After Work

Top-expert-level AIs continue the economic transformation. By early 2036, there are 200 million AIs,[145] equivalent to a workforce of around 100 billion humans, and 2 billion robots with some mixture of humanoid and other specialized form factors.[146] The AIs think faster than humans, and the robots work harder.[147] Any task that was previously bottlenecked by cognitive or physical labor gets sped up until some new bottleneck is encountered. Many previously unresolvable bottlenecks have fallen to armies of genius-level intellects thinking hard about how to resolve them.



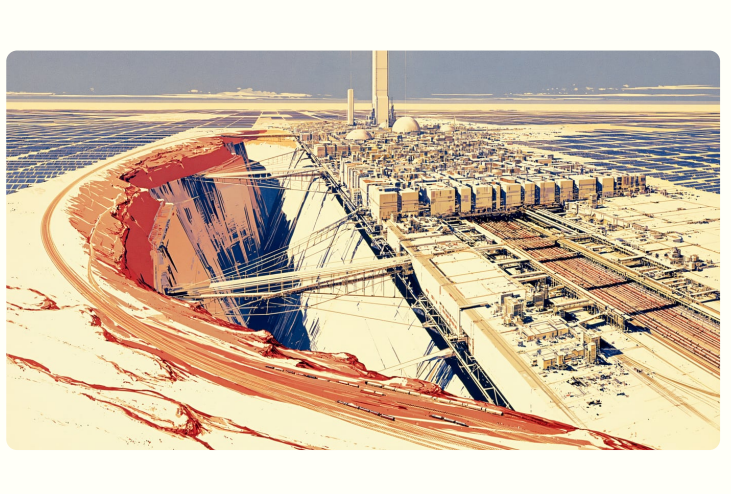
This figure shows the population of AI and robots in human-equivalent labor.[148]

Human, AI & Robot Population – ai-2040.com

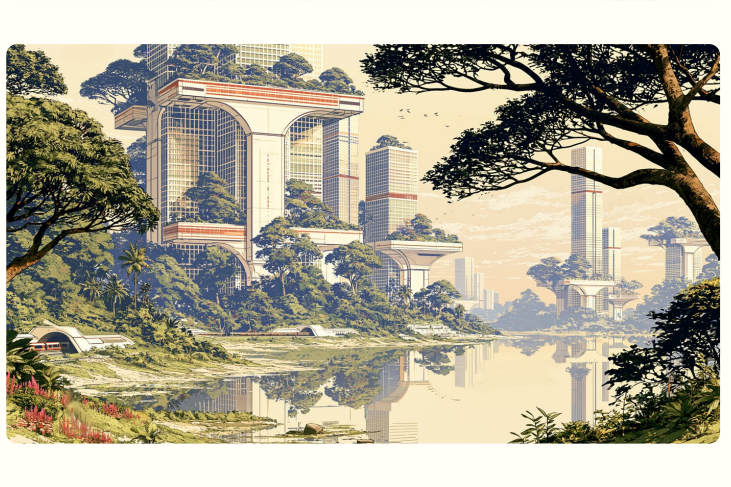
The economy has been rebuilt from the ground up. The fraction of tasks automated has gone from "a sizable chunk of cognitive labor and basically none of physical labor" in 2030, to now "pretty much everything."

Many of the robots work on building more robots. But there are plenty of other tasks to be done: constructing shipyards to build more data tankers, tiling deserts with solar farms, replacing humans at service jobs. In states that deregulate housing quickly enough, they're busy building skyscrapers: land prices are soaring, and governments that have tried everything else grudgingly give in to the YIMBYs and go vertical.[149]

The world is basically being divided into three kinds of territory:



Industrial Special Economic Zones: Picture a gigantic strip mine—an artificial Grand Canyon—next to a city-sized factory full of robots and empty of humans.



Arcologies: Picture a tall skyscraper-mall complex surrounded by nature. Good weather, close to beaches and other cities, but not close enough to be blocked by zoning regulations.

Historic & Nature Preserves: Everything else, i.e., 99% of the world. Yosemite, Paris, SF, New York—these places look basically the same as they did in 2025, or 1995 for that matter. A lot more tourists, though.

▶ *The AI and robot economy*

When the Citizen's Dividend was first passed, people found it shameful to quit their job and live off government largesse. But the changing economic situation steamrolled over the stigma: By now, only 26% of Americans have jobs.

What is life like for the majority of Americans now living off their dividend?

Many of the world's evils have dramatically reduced. Malnutrition, lack of medicine, and homelessness are nearly banished. Many diseases have been cured. Crime rates are lower than ever before.

Meanwhile, most of the good things in life continue. For example, finding romance, or raising a family. People also find meaning in hobbies and competitions, and in learning and trying new things. People are wealthier and have more free time, plus there are new technologies that help, like AI matchmakers, better medicine, and AI tutors.[154] And of course even the poorest people can now afford exotic vacations, amazing games, and enthralling entertainment.

People used to get meaning from feeling like they were useful, like they were contributing something to society. That feeling is harder to come by these days.

But it's not completely gone. There are still problems in the world, and you can still contribute to solving them, partly by donating or volunteering but primarily by being politically active.[155] Your vote is your most important asset.

The honest AI forecasters are helpful here. Years ago, if an AI said that one presidential candidate was better than another, people would suspect bias and the embarrassed company would retrain the AI to evade such questions. Now, thanks to transparency and improved alignment techniques, there are much smarter AIs that people can *see* don't have any biases trained into them,[156] that have built up an excellent track record over several years. When *they* weigh in on policy questions, people listen, especially when different AIs trained by different companies converge to the same answer.

These AIs say that normal people no longer have significant *economic* leverage over the future; human labor is largely obsolete. This puts their *political* leverage in jeopardy.[157] There's still a chance that things will trend towards techno-oligarchy as corporations, politicians, and wealthy shareholders get in bed with each other and gradually disempower the common folk.

But for the first time, enough people have enough freedom from the daily struggle to think about the situation deeply, and the tools to chart it clearly and work through the implications. So the quality of political discourse improves. During the 2036 election season, voters are unprecedentedly well-informed, and the politicians that win have a genuine commitment to responsible stewardship of the upcoming singularity.

2037: The Apocalyptic Arrival of Truth on Earth

For several years now, a hundred million top-expert-level AIs have been running at 100x human speed.[158] This causes scientific progress to go 10x-1000x faster than without AIs, depending on the field.[159]

So humanity has been learning many, many new things.

Including things that lots of people really don't want to believe. And other things that were supposed to stay secret.

Great paradigm shifts in science used to take decades to play out, as experiments were conducted, papers written, and conferences convened. Now these revolutions play out in months.[160] Everyone is grateful for the new disease cures and cheap clean energy, but in other ways this explosion of ideas is profoundly destabilizing. Just as the scientific and economic changes of the nineteenth century produced new ideologies like Marxism and Social Darwinism, and breathed new life into old ideas like atheism, so these new paradigm shifts have unpredictable downstream effects on the ideologies of the day.[161] Political coalitions and fault lines have dissolved, and new ones have formed and dissolved and formed again.

There are also new social technologies. For example, it is now cheap to spin up a team of AIs as good as the world's best historians and private investigators, except faster and with access to new forensic tools. Many skeletons in closets are revealed.

Privacy-preserving auditing is now well-established. Multiple trusted third parties transparently train auditor agents; these agents can be plugged into a trove of private data, answer questions about what they see, and then get deleted. It took a while for adoption to be widespread, but now it's normal for politicians to say "The accusations are false, and to prove it, I'll open up my entire trove of personal data to privacy-preserving inspection; just ask the auditor agents if anything seems to support the allegations and if any data appears to be missing or faked."

The result is a somewhat better breed of politicians[162] and, equally importantly, an *increase* in the ability of states to verify compliance with laws and deals while simultaneously *decreasing* the amount of invasive surveillance. Voters ask: Now that privacy-preserving auditing is a thing, why does anyone in the government need to be able to see our data?

Then lie detectors arrive and turbocharge this.[163] For decades, polygraphs were mostly security theater. But now, with the help of AI, lie detectors are starting to actually work quite well.

In another world, this could have been disastrous. Leaders of AI projects could have used them to purge potential whistleblowers; leaders of governments with frontier AI projects could have used them on a grander scale to become dictators. The nightmare scenario would be a world where lie detectors are used *by* the powerful but not *on* the powerful.[164]

But in this world, lie detectors turn out to be a blessing, because there are many frontier AI projects spread out over many different countries. It's obvious to everyone that insofar as lie detectors are feasible, they'll soon be independently invented and become widely available via diverse third party providers. Politicians and CEOs can still try to use them to purge the disloyal, but they in turn would be purged by voters and shareholders demanding they say 'under oath' that they haven't done such a thing. In fact, anticipating this possibility years ago, powerful people have generally been acting more responsibly. A few sociopaths retired gracefully to get out of the spotlight; others clung to power and now go down in scandal.

There is a constantly-evolving discourse about when it's appropriate to ask someone to say something 'under oath.' Politicians are under the most scrutiny, but even they can usually get away with saying "I won't answer that question about my private life, that's none of your business." They can't get away with "How dare you ask me whether I plan to uphold my campaign promises."[165]

▸ *Should lie detectors be allowed, banned, or regulated?*

International agreements are now more stable than ever, because it's so hard to cheat. Anyone deciding to cheat needs an excuse for not being willing to prove that they aren't cheating. The traditional excuse was "we can't prove it without revealing important state secrets, sorry" but now technology exists to answer questions like "are you cheating on this deal" without leaking any other information. If any government-approved covert AI projects existed, they would have been discovered by now.

2038: AI Alignment Is Now a Science

The AI alignment situation has vastly improved since the mid-2030s. There is a mature science of goal, drive, and value formation in artificial neural nets. Want your new AI to be honest? There's standard protocol for training true honesty (as opposed to too-clever-to-get-caught lying or self-deception). How do we know it works? Because there's a textbook explaining the theory behind why it *should* work, plus literature on various alternative theories that were proposed and disproven. Also because new interpretability tools let us directly observe what and how AIs think, and the results confirm the predictions of theory. There are similar protocols for obedience, altruism, and a long and growing list of other desired traits. Unless the alignment science is somehow all wrong, typical AIs are now more virtuous than the most virtuous humans. This alone has profound effects: it's as if saints and angels were walking among us.[167]

▸ *Incentivizing safety work*

The discussion shifts from how to make AIs good to what "goodness" really means. This looks partly like philosophy and partly like case law: which of the million possible definitions of harmlessness or altruism do we want, exactly? How should an AI handle situations where it discovers a philosophical argument for a different kind of harmlessness or altruism, or an important ambiguity in the definition?

Different AIs have different alignment targets programmed in.[173] Some take orders from the CCP, some from the US President, some from different members of Congress, others from politicians in other countries, and many more from other individuals in their roles as advisors or assistants. Other AIs don't take orders from anyone *per se,* but instead work towards various missions. For example there are AI-run corporations that serve the interest of shareholders, and AI-managed nonprofits that serve charitable missions. There are even experimental AI-run courts and police departments, supposedly superhumanly fair and incorruptible.

Most alignment researchers are now confident that current AIs are aligned to their respective targets, and worry about the next steps: if we put these AIs in charge of designing future, much smarter AIs, will that go well? If we put these AIs in charge of more societal institutions and processes, enough that humans could never realistically 'pull the plug,' will that go well?

Already most people think that things will probably go well, but for something as important as this, they want something stronger than 'probably.' So the global pause at top-expert-level continues as discussions and negotiations play out.

2039: Beginning to Trust AIs

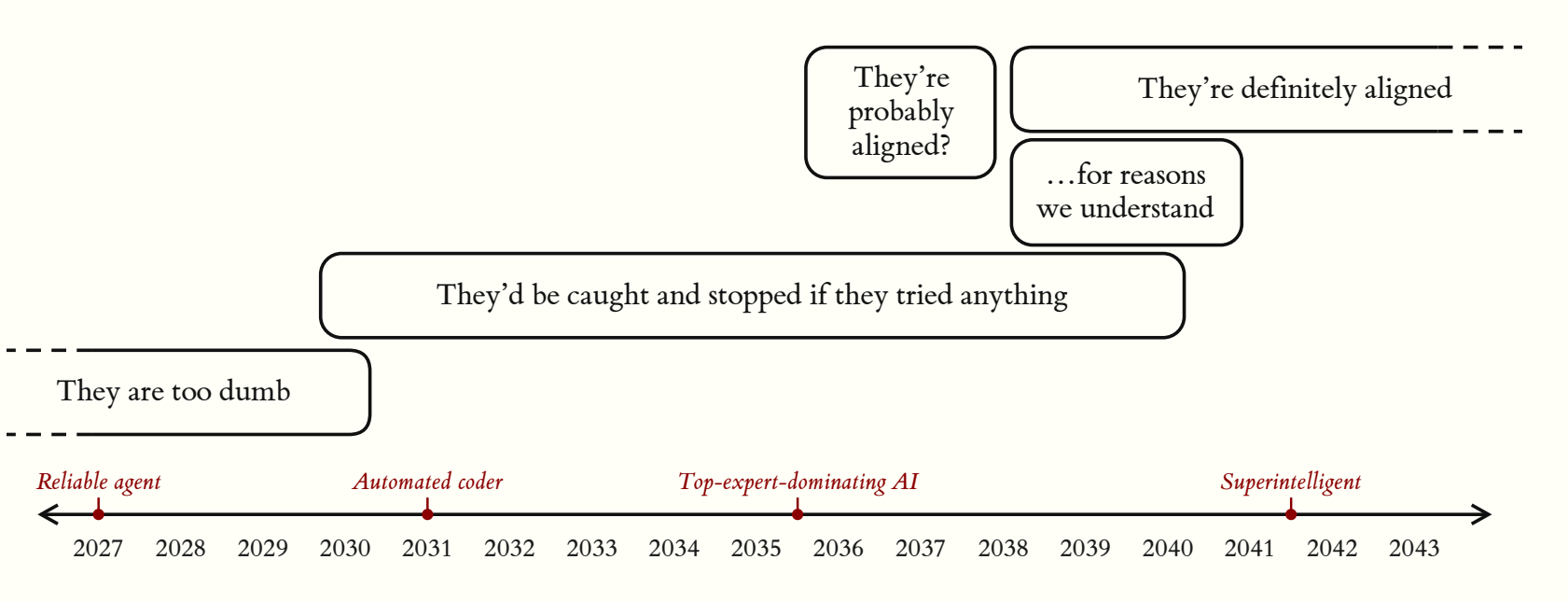
In at least three different ways, the world is negotiating whether and how to dramatically empower AIs.

First, AIs have become increasingly load-bearing in all facets of society, from business to politics to even (some parts of) the armed forces. So far they have been advisors rather than final authorities. Often this is a fig leaf, and the human authorities have rubber-stamped their decisions, but there's always been the option of ignoring their advice and pulling the off switch if something goes wrong.

Getting rid of that option would be, in some cases, really good. By now AIs can be made to be more reliable and virtuous than humans; why *aren't* we putting such AIs in charge? When we sign treaties with other nations, shouldn't we demand that they train an AI sworn to uphold their end of the bargain, and integrate it into their government so that they literally can't cheat?

Relatedly, requiring control-based safety cases is why AI capabilities have mostly stalled at the top-human-expert level. Without that requirement, the AIs could get much, much smarter, and produce a wave of abundance and tech progress that would make the 2030s look like the 2020s. Scaling beyond human level will require relying on alignment based safety cases—arguments that AIs have *robustly* internalized the values they are supposed to have. Each step beyond that will require *deferring* to the previous AIs, trusting their judgment about the safety of the next generation.

▸ *Alignment over time in Plan A*



AI Alignment Eras in Plan A – `ai-2040.com`

Finally, by global agreement, compute and robots have been capped-and-traded since 2032, keeping economic growth to only about 100% per year.[174] Middle powers have grown at roughly the same rate as the US and China, because in 2032 they used their remaining leverage to secure greater shares of global robot and compute production than they would have had without a deal. The Citizen's Dividend is now $10M/yr for Americans (inflation-adjusted!), and even the poorest non-Americans get about $1M.[175] The robot economy is shifting the bulk of activity to space, so that Earth's environment and historic spaces will be protected. As soon as the caps are lifted, off-planet factories will begin churning out more robots and mining equipment with which to construct more factories, and so on. Doubling times for the space economy are projected to be well below one year initially and will drop rapidly insofar as AIs are allowed to improve beyond top-expert level.

The caps are relics of an earlier era when threats like deal collapse loomed larger. Now it's generally agreed that they should be loosened, at least in space, but negotiations are ongoing about the details.

▸ *Why we choose to hand off to AIs in this scenario*

2040: Passing the Torch to AIs

There are still many problems to be solved and issues to be settled, many of which couldn't be anticipated from 2026.

▸ *Examples of problems/issues that still remain*

But civilization is now reasonably well-equipped to handle whatever comes up. The best AIs are aligned, and to publicly-visible values. Power over the best AIs is distributed far more evenly than it was in 2026. Public epistemics are better than ever.

So over the course of the year, regulators around the world loosen the requirements that had been holding back progress.

Gradually more institutions and equipment are handed over to AIs, such that it's no longer true that humans could shut it all down if they wanted to. In fact it's extremely untrue: Soon many of the world's militaries are autonomous, run by AIs sworn to uphold various constitutions and treaties.[176]

AI capabilities are also advancing again. From the AIs' 500x-human-speed perspective, what's happening is a careful scaleup in AI research, under conditions of total research transparency and guardrails negotiated by numerous factions.[177] Soon there will be AIs that are incomprehensibly superintelligent, yet trusted, because we trust the AIs that built them, which we trust because we trust the AIs that built *them*, in a chain all the way back to the AIs of 2040 which we trust because alignment experts thought carefully about the safety cases and concluded they were solid.

Compute and robot caps are tightened on Earth but loosened in space; Earth is to become a preserve, whereas the extraterrestrial economy starts doubling faster and faster.

Even the experts who spent years understanding the latest safety cases can't help but feel nervous—is it all going to go wrong somehow, for reasons we didn't anticipate? Have the AIs been lying to us this whole time, waiting for the right moment to betray? The safety cases show that's impossible, of course…

There is no single moment where humanity relinquishes control. But in theory, there is a point of no return; on some specific day, the AIs are smart enough, and control enough of the world's technological and economic infrastructure, that they really could take over the world if they wanted to. On the most popular operationalization of this question, the forecaster AIs project that this moment will be reached one day in late October. Their 95% confidence interval is months wide, so they are almost certainly wrong about the exact time. Still, people observe the moment in their own ways. Some spend the night in prayer vigils. Others sit at their computer screens, monitoring the situation.

You and your friends throw an End Of The World Party, counting down to the fateful hour with champagne and good company. Some of you are old enough to remember a similar situation at the turn of the millennium, when the Y2K bug and impending new era introduced the phrase "party like it's 1999" to the lexicon; the celebrations of late 2040 are more desperate and correspondingly more festive.

When there is no news by sunrise, you fall into a fitful sleep, dreaming of the ineffable future.

Postscript

We think the world is asleep at the wheel. People say the words "AI will be transformative" without thinking concretely and seriously about the implications of broadly superhuman AI.

AI companies, lobbyists, and think tankers often offer incremental proposals to decrease risk around the edges, but only rarely offer ambitious plans aimed at actually solving the big problems.[178]

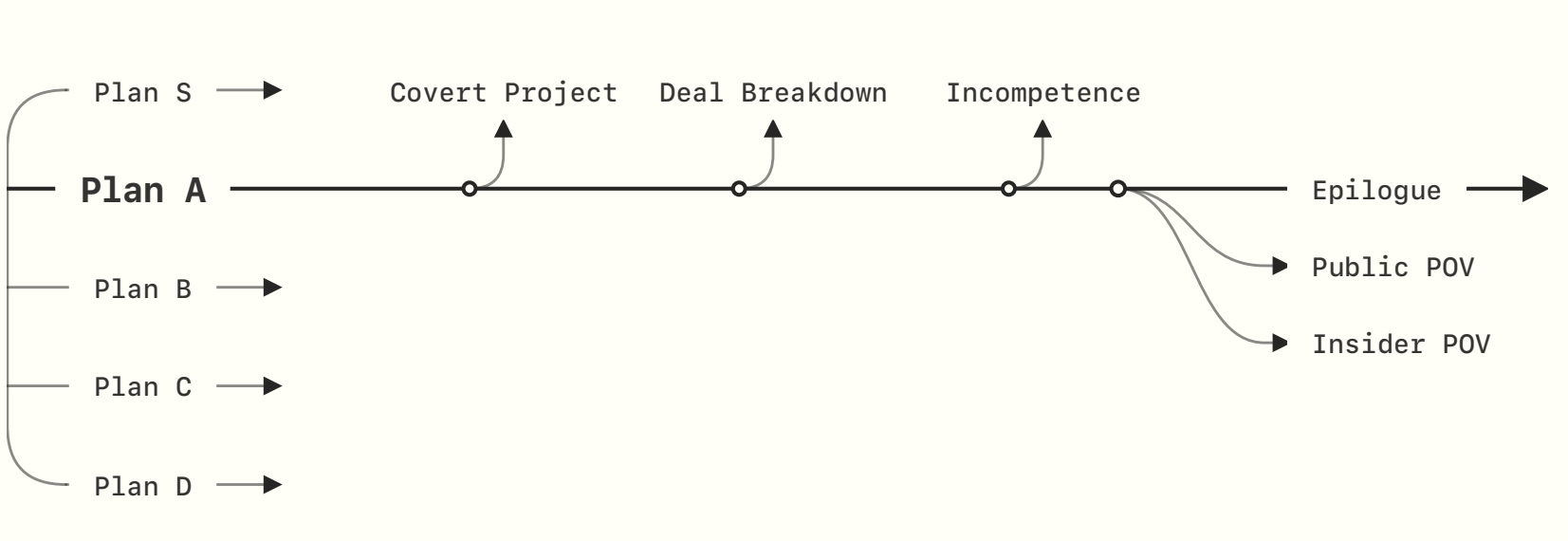
We think that following *only* incremental proposals will lead to a scenario like the race ending of AI 2027, in which misaligned AIs take over.[179] If we get lucky and alignment is easier than expected, then they'll lead to a scenario like the slowdown ending of AI 2027, in which a tiny group of individuals get to decide the shape of the future and could easily become permanent oligarchs if they so choose. Something more ambitious must be done. But what?

Politicians should be banging at the door of the AI companies,[180] asking them for a comprehensive overall plan for how companies and governments should act from now until superintelligence, which should be similarly detailed to Plan A. Then, people should apply scenario scrutiny to these plans, asking questions like "what would it look like to implement them? How long would it take? What would happen next? Who would build superintelligence, and when, and how? What assumptions does success depend on?"

We at the AI Futures Project are attempting to be the change we wish to see in the world. We do so with some trepidation, knowing that a detailed plan has a wider attack surface than one that keeps things conveniently vague. We hope critics will judge us against the existing state-of-the-art for plans to navigate the AI transition (if they can find any) and not against some hazy but pleasant fantasy where no one has to make any hard choices yet everything will probably be fine.

Sometime in the next few years the US government will be in crisis mode, pondering what to do about scarily powerful AIs and blisteringly fast AI progress. We think the choice is not an easy or a simple one. Plan A is our current best guess, but hopefully a better plan will exist before it's too late. We hope that the best ideas from Plan A will be adopted and the worst ideas discarded.

You've reached the end of Plan A



KEEP EXPLORING

Public Perspective
What Plan A feels like to an average citizen

Insider Perspective
What Plan A feels like to an AI researcher

Epilogue
Life after superintelligence

Choose another path
Read another plan scenario